Special Relativity Notes

Tensor Algebra: Symmetry, Contraction, Kronecker Delta and Levi-Civita Tensor

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1 Addition / Subtraction of Tensors

To perform addition/subtraction, the rank of the tensors has to be the same.

Ex:

$$A_i + B_i = C_i$$
$$A^\mu + C^\mu = D^\mu$$
$$M^{\mu\nu} - N^{\mu\nu} = C^{\mu\nu}$$

2 Symmetric and Antisymmetric Tensors

If $A_{ij} = A_{ji}$, then A is a **symmetric tensor**.
If $A_{ij} = -A_{ji}$, then A is an **antisymmetric tensor**.

For $i, j = 1, 2, 3$:

$$A = \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{12} & A_{22} & A_{23} \\ A_{13} & A_{23} & A_{33} \end{pmatrix} \qquad \text{symmetric}$$

$$A = \begin{pmatrix} 0 & A_{12} & A_{13} \\ -A_{12} & 0 & A_{23} \\ -A_{13} & -A_{23} & 0 \end{pmatrix} \qquad \text{anti-symmetric}$$

3 Contraction of Indices

Start from the transformation of a mixed (rank-2) tensor:

$$A^i{}_j \;\rightarrow\; A'^i{}_j = \sum_{r,k} \frac{\partial x'^i}{\partial x_k} \frac{\partial x^r}{\partial x'_j} A^k{}_r$$

If $r = k$, the two partial derivatives combine into a Kronecker delta by the chain rule:

$$A'^i{}_j = \delta^i_j \sum_r A^r{}_r$$

If, further, $i = j$ (summing over the free indices as well):

$$A'^{i}{}_{i} = \sum_{r,k} \delta^{k}_{r}\, A^{r}{}_{k} = \sum_{r} A^{r}{}_{r} \qquad (r = k)$$

The trace $A^{r}{}_{r}$ is invariant under the coordinate transformation — contracting an upper and a lower index produces a lower-rank invariant quantity.

4 Kronecker Delta

Kronecker delta: $\delta_{ij},\ \delta^{ij},\ \delta^{i}_{j}$

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \qquad \text{(tensor of rank 2)}$$

5 Levi-Civita Tensor

$$\text{Levi-Civita tensor} \qquad \epsilon_{ijk} = \begin{cases} 0 & \text{if } i = j = k \ \text{ or } \ i = j \ \text{ or } \ j = k \ \text{ or } \ i = k \\ +1 & \text{if } i \neq j \neq k \ \text{ and in cyclic order} \\ -1 & \text{if } i \neq j \neq k \ \text{ and } \textit{not} \text{ in cyclic order} \end{cases}$$

Ex: $\epsilon_{123} = 1, \qquad \epsilon_{213} = -1, \qquad \epsilon_{312} = 1$

6 Isotropic Tensor

Isotropic tensor: if the components of a tensor remain the same in all rotated frames.

Ex:

$$\delta^{i}_{j} \ \rightarrow\ \delta'^{i}_{j} = \frac{\partial x'^{i}}{\partial x_{r}}\,\frac{\partial x^{k}}{\partial x'_{j}}\,\delta^{r}_{k} = \frac{\partial x'^{i}}{\partial x'_{j}} = \delta^{i}_{j}$$

$$\delta'^{i}_{j} \ \rightarrow\ \delta''^{i}_{j} = \delta^{i}_{j}$$

7 Outer Product

$$A_i \otimes B^j = C_i{}^j$$

$$A_i \otimes B_j = C_{ij}$$

$$B_{\mu\nu} \otimes C^{\alpha} = D_{\mu\nu}{}^{\alpha}$$

low rank tensor $\longrightarrow$ high rank tensor